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Prediction of Retinopathy of Prematurity and Treatment in Very Low Birth Weight Infants Using Machine Learning on Nationwide Non-Imaging Clinical Data

Jae Kyoon Hwang[a] Donggoo Jung[b] Hyun-Kyung Park[a] Daehyun Kim[b] Hyun Jeong Do[c] Seong Hee Oh[c] Seung Hyun Kim[d] Tae Hyun Kim[e] Hyunseung Jin[c]

[a]Department of Pediatrics, Hanyang University College of Medicine, Seoul, South Korea; [b]Department of Artificial Intelligence, Hanyang University, Seoul, South Korea; [c]Department of Pediatrics, Gangneung Asan Hospital, University of Ulsan College of Medicine, Gangneung, South Korea; [d]Department of Pediatrics, Samsung Medical Center, Sungkyunkwan University School of Medicine, Seoul, South Korea; [e]Department of Computer Science, Hanyang University, Seoul, South Korea

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Keywords
Retinopathy of prematurity · Machine learning · Very low birth weight infants · Multilayer Perceptron · Neural Oblivious Decision Ensembles

Abstract

Introduction: Retinopathy of prematurity (ROP) remains a leading cause of preventable blindness in preterm infants. This study aimed to develop machine learning (ML) models using non-imaging clinical data to predict ROP, severe ROP (sROP), and treated ROP (tROP) in very low birth weight (VLBW) infants. ***Methods:*** We utilized nationwide clinical data from the Korean Neonatal Network, including 44 perinatal and neonatal variables. Two deep learning models, Multilayer Perceptron (MLP) and Neural Oblivious Decision Ensembles (NODE), optimized for tabular data, were applied. Additionally, we developed simplified models using eight key variables selected through clinical and algorithmic relevance. ***Results:*** MLP and NODE models demonstrated high predictive performance. For the full 44-variable models, the area under the receiver operating characteristic curve (AUROC) was as follows: ROP (0.853/0.855), sROP (0.888/0.890), and tROP (0.905/0.909). The reduced 8-variable models yielded comparable AUROCs: ROP (0.851/0.855), sROP (0.895/0.895), and tROP (0.910/0.909). ***Conclusion:*** The proposed ML models based on nationwide non-imaging clinical data enable early risk identification and timely intervention for ROP in VLBW infants. This cost-effective and scalable approach may help improve outcomes, especially in resource-limited settings.

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Plain Language Summary

Retinopathy of prematurity (ROP) is an eye condition that can affect premature babies (babies born too early, before 37 weeks of pregnancy). In ROP, abnormal blood vessels

Jae Kyoon Hwang, Hyun-Kyung Park, and Donggoo Jung contributed equally to this work.

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Correspondence to:
Tae Hyun Kim, taehyunkim@hanyang.ac.kr
Hyunseung Jin, solchani72@ulsan.ac.kr

grow in the retina, which can lead to vision problems or even blindness. To prevent serious outcomes, early detection and treatment are essential. However, not all hospitals have enough trained eye specialists to screen every baby at risk. For this reason, this study aimed to develop an easier way to identify babies who may need eye examinations using commonly collected medical data. To address this goal, the researchers analyzed health records of premature babies collected across South Korea. Using a method called machine learning, which allows computers to find patterns in data, they created two computer models. These models could predict which babies were more likely to develop severe forms of ROP or need treatment. Importantly, the models used only basic clinical information like birth weight, oxygen support, and medical complications, without requiring eye images. The models showed high accuracy even when using just a few key factors. By identifying risk in this way, this type of model can help hospitals recognize high-risk babies early and refer them for specialized care, even if eye doctors are not available on site. It offers a practical, low-cost tool for improving ROP screening programs, especially in areas with limited resources.
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Introduction

Retinopathy of prematurity (ROP) is one of the serious complications of prematurity and is a disease that can lead to vision damage or blindness [1]. The survival of preterm infants has increased in the last few decades owing to rapid improvements in neonatal intensive care. Consequently, the incidence of ROP has increased, particularly in newly industrialized countries, comprising a "third epidemic" [2–4]. Accordingly, many efforts, including screening tests, have been made to detect and treat ROP early, especially in newly industrialized countries that lack the facilities and manpower to diagnose and treat ROP.

The frequency of ROP tends to increase with lower gestational age and birth weight, similar to other complications of prematurity [5]. In addition, several studies have reported associations with risk factors, such as duration of ventilator use, antenatal steroid use, surfactant use, erythrocyte transfusion, severe clinical score, poor nutrition, hyperglycemia, sepsis, and bronchopulmonary dysplasia [4, 6–8]. Recently, associations with smooth weight gain, and IGF-1 and IGFBP-3 levels have been identified, and prediction methods such as the ROPScore, CHOP-ROP, and WINROP, using a scoring system that combines the trends of gestational age, birth weight, and weight gain, have been limited to use in the clinical setting [9–11].

Artificial intelligence (AI) technologies have recently been used to solve various medical problems, including diagnosing diseases, choosing treatments, and predicting the risks and progression of diseases. Machine learning (ML) algorithms, specifically artificial neural networks (ANNs), allow a more accurate estimation of complex patterns by learning from training data and can predict which patients will develop a disease as an early and supportive diagnostic tool for clinicians. Numerous studies have been conducted on ROP; however, most have focused on retinal image analysis [12, 13]. To the best of our knowledge, no studies have been published on ML models employing tabular ROP data, comprised nationwide prospective web-based real-time demographic and clinical information. In this study, we apply Multilayer Perceptron (MLP) [14] and Neural Oblivious Decision Ensembles (NODE) [15] methods. MLP has a strong ability to learn complex nonlinear relationships, whereas NODE is optimized for tabular data, offering improved interpretability and ensuring stable performance. Furthermore, with the collection of additional datasets in the future, deep learning models can achieve even stronger performance and enhanced interpretability. This study aimed to develop ML prediction models for the occurrence of ROP and its severe stages, as well as the need for treatment, using non-imaging demographic data from a nationwide cohort, thereby making it beneficial for regions with limited access to fundus cameras or ophthalmology personnel.

Methods

Data Collection

We obtained data from infants registered in the Korean Neonatal Network (KNN), a nationwide prospective cohort registry of very low birth weight (VLBW) infants that accounts for more than 70% of the total admissions of VLBW infants born in Korea [16]. This study was approved by the Institutional Review Board of GangNeung Asan Hospital (2021-12-014). The Institutional Review Board approved the data registration in the KNN at each participating hospital.

The cohort comprised 19,980 infants born between January 5, 2013, and December 31, 2022, weighing <1,500 g. We excluded infants with major congenital anomalies, those without ROP examinations, and those with missing outcome information, resulting in 13,349 infants with available outcomes for analysis (online

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suppl. Fig. 1, Methods A; for all online suppl. material, see https://doi.org/10.1159/000550513). Three analyses were conducted. For ROP prediction, the prediction rate was analyzed for the target group with ROP occurrence (ROP prediction), the severe ROP (sROP) occurrence group with ROP stage 3 or higher (sROP prediction), and the target group requiring treatment (tROP prediction) such as surgery or anti-vascular endothelial factor (VEGF) treatment.

Disease Definitions

The manual of operation of the KNN guided the definitions of the collected data (see online suppl. material). Ophthalmic screening for ROP followed an established guideline set by several pediatric and ophthalmologic organizations that recommends screening for ROP in infants born with body weights less than 1,500 g or gestational age less than 32 weeks [17]. All VLBW infants underwent indirect binocular ophthalmoscopy performed by experienced ophthalmologists. The ROP stages were classified from 1 to 5 according to the International Classification of ROP [18], with sROP defined as stage ≥3. The highest stage before discharge was recorded. Treatment decisions, including anti-VEGF injection or surgical interventions (cryotherapy, laser photocoagulation, vitrectomy), were made per each center's ophthalmological policy. From approximately 120 registry items, 44 variables available before the first ROP screening window were selected based on temporal availability, clinical relevance, and data completeness (online suppl. Table 1, online suppl. Methods B).

Statistical Analysis

Baseline characteristics between infants with and without ROP were compared using Student's t test for continuous variables and the chi-square test for categorical variables, with significance set at $p < 0.05$ (Table 1).

Data Preprocessing

The dataset (n = 13,349) was randomly divided into training, validation, and test sets (Table 2). Missing values were handled using an iterative imputation method based on BayesianRidge [19] regression, considering correlations among features.

Model Selection and Implementation

We employed two models for our analysis: model 1, a conventional MLP, and model 2, which utilized NODE. Unlike typical neural networks, NODE is specifically designed for structured, tabular data. It incorporates oblivious decision trees (ODTs) into a differentiable neural architecture, enabling gradient-based optimization. By combining the interpretability of decision trees with the flexibility of deep learning, NODE provides a robust alternative to conventional models, particularly in datasets lacking spatial or sequential structure.

Training

We used the binary cross-entropy loss function to train the proposed deep learning models (MLP and NODE) separately to predict the ROP, sROP, and tROP groups. For loss minimization, we used the RMSprop optimizer with learning rates determined via a learning rate finder technique [20] selecting rates from regions of most rapid loss reduction to ensure efficient convergence. To mitigate overfitting, we applied dropout (0.5 for model 1, 0.1 for model 2) and employed early stopping, halting training if the loss failed to improve for 20 consecutive iterations.

Model 1 (MLP) was trained with a batch size of 32, while model 2 (NODE) utilized task-specific configurations of six layers, eight trees, and varying tree depths (12 for ROP prediction, 4 for sROP, and 6 for tROP) to optimize performance. Deep learning models were implemented in PyTorch [21], with conventional ML analyses conducted using scikit-learn [22].

Results

Study Population and Baseline Characteristics

A total of 8,949 patients were classified as having no ROP (ROP stage 0), 1,388 as having stage 1, 1,415 as having stage 2, 1,569 as having stage 3, 25 as having stage 4, and 3 in stage 5 (online suppl. Fig. 1). Table 1 presents the characteristics of the groups with and without ROP and highlights several representative factors. There were 1,597 infants with sROP (stage ≥3) and 1,630 infants with treated ROP (tROP), a group that included those who had undergone operation or received anti-VEGF injection.

Proposed ML Algorithms

In this study, we used several deep neural networks to predict ROP in infants. The networks were designed to take 44-dimensional input vectors, each producing a single output value for binary classification tasks. We also explored a streamlined approach focusing on eight key factors essential for prediction, ensuring that the model remains efficient while maintaining predictive accuracy.

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Table 1. Representative clinical characteristics of infants

ROP	Stage 0 (*n* = 8,949)	Stage 1 (*n* = 1,388)	Stage 2 (*n* = 1,415)	Stage 3 (*n* = 1,569)	Stage 4 (*n* = 25)	Stage 5 (*n* = 3)	*p* value (no ROP vs. any ROP)
Gestational age (mean±SD), weeks	30.2±2.4	28.2±2.1	26.9±2.0	26.0±1.8	25.3±1.9	28.3±3.9	<0.001
Birth weight (mean±SD), g	1,198.2±219.3	1,050.0±244.0	900.0±235.0	799.2±213.2	754.7±252.9	790.0±91.7	<0.001
Male, *n* (%)	4,440 (49.6)	702 (50.9)	700 (49.5)	789 (50.3)	15 (60.0)	1 (33.3)	0.318
Maternal age (mean±SD), years	33.5±4.2	33.5±4.2	33.6±4.3	33.6±4.2	33.9±4.4	30.7±4.2	0.369
PROM, *n* (%)	2,886 (32.4)	535 (38,5)	551 (38.9)	617 (39.3)	7 (28.0)	1 (33.3)	<0.001
Antenatal steroid use, *n* (%)	7,712 (86.2)	1,230 (88.6)	1,239 (87.6)	1,406 (89.6)	23 (92.0)	3 (100.0)	<0.001
Vaginal delivery, *n* (%)	1,539 (17.2)	265 (19.1)	306 (21.6)	335 (21.4)	6 (24.0)	0 (0.0)	<0.001
Resuscitation at birth, *n* (%)	7,751 (86.6)	1,320 (95.1)	1,381 (97.6)	1,545 (98.5)	25 (100.0)	3 (100.0)	<0.001
RDS, *n* (%)	5,920 (66.2)	1,211 (87.3)	1,337 (94.5)	1,518 (96.8)	25 (100.0)	2 (66.7)	<0.001
Air leak syndrome, *n* (%)	167 (1.9)	43 (3.1)	87 (6.2)	112 (7.1)	2 (8.0)	1 (33.3)	<0.001
Pulmonary hypertension, *n* (%)	314 (3.5)	100 (7.2)	196 (13.9)	291 (18.6)	10 (40.0)	1 (33.3)	<0.001
Postnatal systemic steroid use, *n* (%)	570 (6.4)	206 (14.8)	365 (25.8)	512 (32.6)	11 (44.0)	1 (33.3)	<0.001
Invasive mechanical ventilation (mean±SD), days	6.7±17.2	15.6±25.4	29.2±32.9	43.7±39.5	80.5±82.7	121.0±150.8	<0.001
Assisted ventilation (mean±SD), days	24.4±27.5	42.3±34.9	63.7±42.3	82.1±49.3	131.8±89.6	135.0±135.1	<0.001
Total oxygen treatment (mean±SD), days	29.2±30.6	51.06±39.1	75.5±47.8	96.5±52.8	137.4±89.1	138.7±131.4	<0.001
IVH (grade ≥2), *n* (%)	726 (8.12)	194 (13.98)	331 (23.39)	483 (30.78)	9 (36.0)	2 (66.67)	<0.001
Sepsis, *n* (%)	1,085 (12.1)	272 (20.0)	393 (27.8)	566 (36.1)	12 (48.0)	1 (33.3)	<0.001
PDA on medication, *n* (%)	2,154 (24.1)	545 (39.3)	652 (46.1)	819 (52.2)	14 (56.0)	2 (66.7)	<0.001
NEC (stage ≥2), *n* (%)	324 (3.6)	79 (5.7)	119 (8.4)	197 (12.6)	7 (28.0)	0 (0.0)	<0.001
TPN duration (mean±SD), days	19.5±18.5	29.5±24.6	41.9±33.3	52.7±39.2	76.7±62.0	97.0±70.2	<0.001
Erythrocyte transfusion, *n* (%)	4,344 (48.5)	933 (67.2)	1,204 (85.1)	1,473 (93.9)	23 (92.0)	3 (100.0)	<0.001
Treated ROP, *n* (%)	0 (0.0)	28 (2.0)	343 (24.2)	1,232 (78,5)	25 (100.0)	2 (66.7)	
Operation	0 (0.0)	21 (1.5)	248 (17.5)	892 (56.9)	25 (100.0)	1 (33.3)	
Anti-VEGF injection	0 (0.0)	12 (0.9)	155 (11.0)	565 (36.0)	11 (44.0)	2 (66.7)	

IVH, intraventricular hemorrhage; NEC, necrotizing enterocolitis; PDA, patent ductus arteriosus; PROM, premature rupture of membranes; RDS, respiratory distress syndrome; ROP, retinopathy of prematurity; TPN, total parenteral nutrition; VEGF, vascular endothelial growth factor.

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Table 2. Size of training and evaluation datasets

Group	Training dataset (80%)				Test dataset (20%)		Total
	training set (90%)		validation set (10%)				
	negative	positive	negative	positive	negative	positive	
ROP prediction	6,405	3,206	702	366	1,842	828	13,349
sROP prediction	8,433	1,178	933	135	2,386	284	
tROP prediction	8,408	1,203	931	137	2,380	290	

ROP, retinopathy of prematurity; sROP, severe ROP; tROP, treated ROP.

Specifically, we utilized MLP [14] to highlight the advantages of deep neural networks. In general, ANNs can achieve improved performance by simply adding layers and enhancing nonlinearity. However, overfitting can occur when the training dataset is limited. We incorporated batch normalization [23] and dropout [24] techniques to address this issue and promote stable training.

Figure 1 illustrates the architecture of the proposed approach. Model 1 (MLP-based) is a binary classifier composed of three hidden layers, each consisting of a block with an activation function, batch normalization, and dropout. This model uses a 44-dimensional or 8-dimensional vector as the input and outputs feature vectors after passing through all layers.

To further enhance the performance, particularly for tabular data, we also employed deep-learning models specialized for structured datasets (model 2). We used NODE [15], a deep learning model specifically designed for tabular data. As shown in Figure 1b, it utilizes ODTs, in which all internal nodes within a tree evaluate the same feature with the same threshold. This structure simplifies the computation and enhances the efficiency for tabular data. Multiple ODTs are trained together in an ensemble, allowing the model to effectively capture complex data patterns.

Performance of Classic ML Models and Proposed Deep Learning Models

The prediction results provided in Table 3 and Figure 2 compare the traditional ML models with our neural ANN approaches (models 1 and 2). The proposed neural methods outperformed traditional learning-based methods in all prediction tasks (ROP, sROP, and tROP prediction) in terms of accuracy and area under the receiver operating characteristics (AUROC). Models 1 (MLP) and 2 (NODE) achieved the highest AUROC scores for predicting ROP [0.853/0.855], sROP [0.888/0.890], and tROP [0.910/0.909]. In our study, model 1 demonstrated a strong performance; however, model 2 surpassed it in tROP prediction. This outcome indicates that model 2 not only preserves feature interpretability, which is critical for sensitive predictions such as the ROP, but also serves as a viable alternative to conventional deep learning models. A detailed analysis of the precision, sensitivity, and F1 score for each task is presented in online supplementary Table 2.

Significant Risk Factors as Analyzed Using SHAP

After producing the ML models (NODE) for ROP prediction, we sorted the top 20 variables that contributed the most to outcome prediction. We used the Shapley values for the contribution analysis and SHAP [25] as the analysis method. Figure 3 shows the importance matrix and SHAP summary plots for the ML models (Model 2) across different tasks. The principal variables that significantly influenced ROP, sROP, and tROP were gestational age, birth weight, days of total oxygen treatment, and days of assisted ventilation. Intraventricular hemorrhage and days of total parenteral nutrition were also identified as important risk factors. Finally, we determined eight major factors (gestational age, birth weight, antenatal steroid use, days of invasive mechanical ventilation, days of assisted ventilation, days of total oxygen treatment, days of total parenteral nutrition, and erythrocyte transfusion) by analyzing the results of the above SHAP and factors known in previous studies to select factors more concisely for clinical application.

Performance comparison between 44 Factors and 8 Factors

The performance of the prediction model using the 44 clinical variables was compared to that of the prediction model using the eight major factors mentioned above (Table 3; Fig. 2). The proposed method demonstrated

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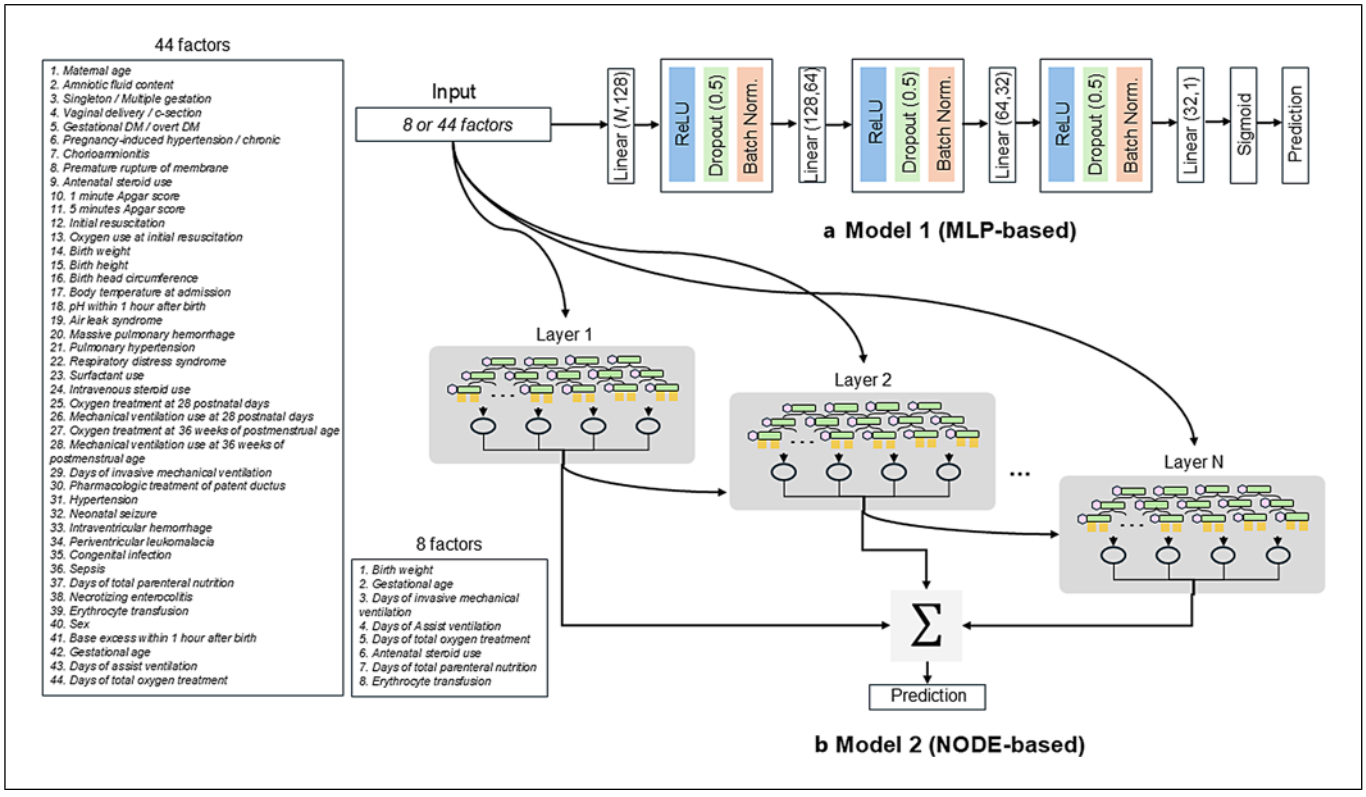


Fig. 1. The architecture of ML models. **a** Model 1 is a 3-layered neural network based on the MLP architecture. **b** Model 2 is based on NODE, where multiple ODTs are combined as an ensemble to make the final prediction. Each model takes 44 or 8 factors as input for ROP prediction. ROP, retinopathy of prematurity; ODTs, oblivious decision trees; NODE, Neural Oblivious Decision Ensemble; MLP, Multilayer Perceptron.

Table 3. Model performance results for each group, measured by accuracy and AUROC

Group	ROP prediction (no ROP vs. any ROP)				sROP prediction (ROP 0, 1, 2 vs. severe ROP)				tROP prediction (untreated vs. treated ROP)			
accuracy/AUROC												
No. of input factors	44		8		44		8		44		8	
Logistic regression	0.703	0.766	0.690	0.713	0.894	0.721	0.894	0.478	0.891	0.728	0.891	0.467
Linear SVM	0.772	0.812	0.807	0.830	0.893	0.881	0.895	0.890	0.891	0.499	0.901	0.904
Radial SVM	0.690	0.535	0.804	0.833	0.894	0.523	0.897	0.830	0.893	0.656	0.905	0.858
K-NN	0.706	0.651	0.748	0.763	0.881	0.579	0.887	0.723	0.880	0.590	0.893	0.780
Decision tree	0.721	0.690	0.720	0.690	0.853	0.656	0.857	0.650	0.852	0.676	0.863	0.695
XGBoost	0.803	0.851	0.798	0.839	0.889	0.884	0.886	0.880	0.898	0.903	0.899	0.899
AdaBoost	0.806	0.850	0.806	0.846	0.892	0.888	0.893	0.891	0.902	0.902	0.900	0.906
Model 1 (MLP)	**0.816**	0.853	**0.816**	0.851	0.896	0.888	0.898	0.895	0.905	0.905	**0.908**	**0.910**
Model 2 (NODE)	0.814	**0.855**	0.815	**0.855**	**0.901**	**0.890**	**0.899**	**0.895**	**0.906**	**0.909**	0.906	0.909

AUROC, area under the receiver operating characteristic curve; K-NN, K-nearest neighbor; ML, machine learning; MLP, Multilayer Perceptron; NODE, Neural Oblivious Decision Ensembles; ROP, retinopathy of prematurity; sROP, severe ROP; SVM, support vector machine; tROP, treated ROP; XGBoost, extreme gradient boosting.

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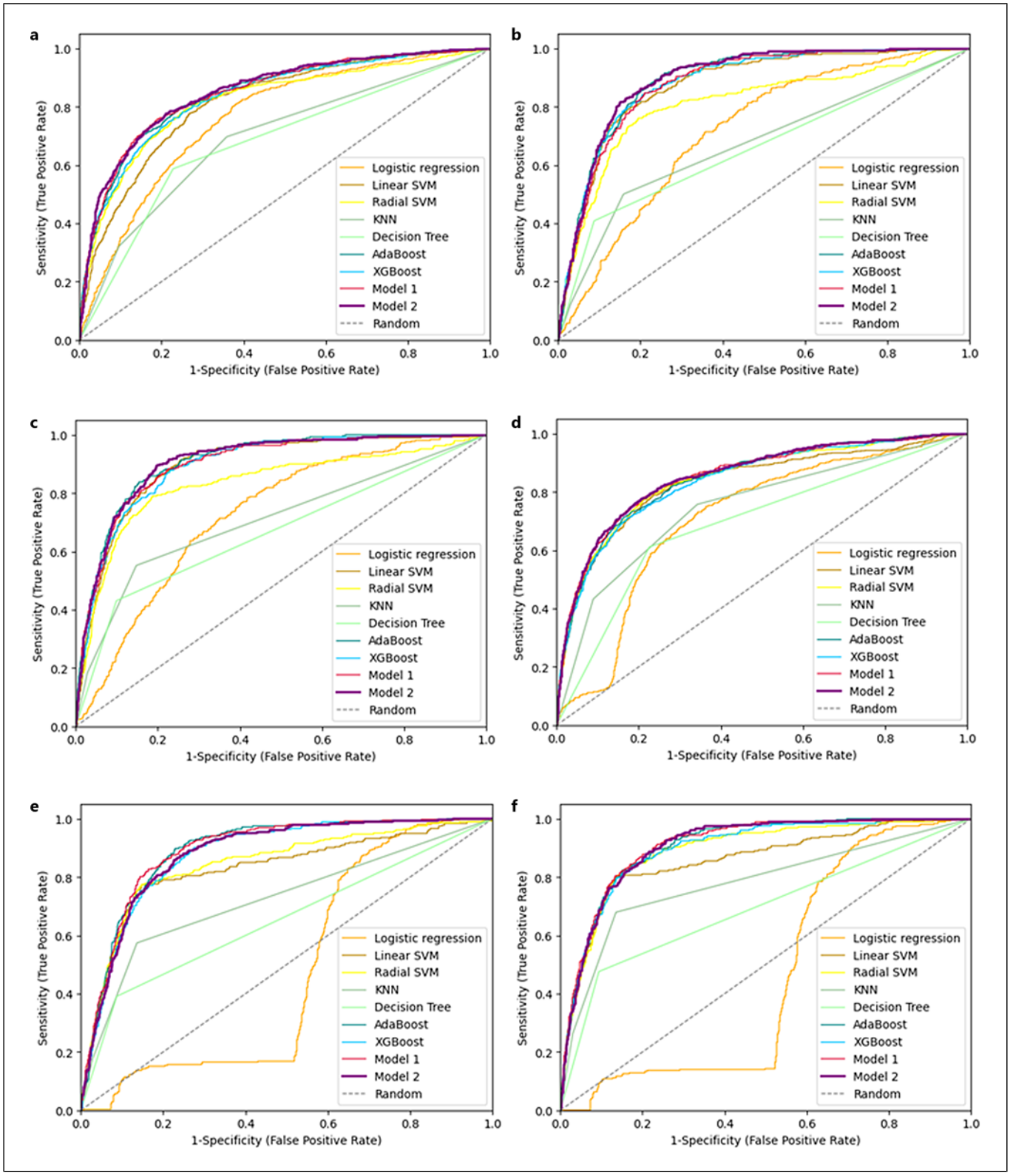


Fig. 2. ROC curves in each prediction model. **a** ROP prediction (44 factors). **b** sROP prediction (44 factors). **c** tROP prediction (44 factors). **d** ROP prediction (8 factors). **e** sROP prediction (8 factors). **f** tROP prediction (8 factors). ROC, receiver operating characteristics; ROP, retinopathy of prematurity; sROP, severe ROP; tROP, treated ROP.

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Fig. 3. Importance of clinical variables contributing to prediction in each group. **a** SHAP-based feature importance plot of ROP prediction. **b** SHAP-based feature importance plot of sROP prediction. **c** SHAP-based feature importance plot of tROP prediction. ROP, ROP, retinopathy of prematurity; sROP, severe ROP; tROP, treated ROP; SHAP, SHapley Additive exPlanations.

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robust performance even when using the eight clinically significant factors. Model 2 (NODE) showed similar AUROC values when using 44 or eight factors (ROP: 0.815 vs. 0.855, sROP: 0.899 vs. 0.895, and tROP prediction: 0.906 vs. 0.909, respectively), suggesting that reducing the number of input variables did not significantly affect predictive performance.

Discussion

In this study, we used ML techniques to predict the ROP, sROP, and tROP in a nationwide registry of VLBW infants in Korea. Our results showed that the MLP and NODE models outperformed the existing predictive models. This trend was consistent in analyses utilizing all 44 clinical variables and in analyses restricted to a subset of eight variables selected based on their significance in the SHAP analysis and their established associations in previous studies. The consistency of these results across the full and reduced variable sets highlights the robustness of AI-based models for predicting ROP risk. This suggests that AI-based models can improve the accuracy of ROP risk stratification even when applied to a limited set of clinically established predictors. Previous studies have focused primarily on predicting the occurrence of ROP. However, our study developed models that predicted the occurrence of ROP and the likelihood of sROP and ROP requiring treatment and showed strong performance in all cases.

In previous studies, prediction models such as the ROPScore, CHOP-ROP, and WINROP have been developed [9–11], which evaluate the possibility of ROP occurrence based on gestational age, birth weight, and weight gain trends. However, these models have shown limitations in their predictive power in various neonatal populations, and there is a need for data on weight gain. Recently, AI-based research has been actively conducted [12, 13], but most has focused on ROP diagnosis using retinal image analysis, and ROP prediction using only clinical data is relatively lacking. In this study, ROP, sROP, and tROP were predicted by applying an ML model using only clinical data. It showed that an AI-based approach could effectively screen high-risk groups for ROP without using retinal images. This is one of the greatest strengths of this study and suggests that it can be used as an alternative screening tool, especially in areas where specialists are in short supply.

In this study, the predictive performance of the model using all 44 variables was not significantly different from that of the model using only eight major risk factors. This suggests that key factors, such as gestational age, birth weight, total oxygen therapy days, and number of days of assisted ventilation, play important roles in the occurrence and severity of ROP. Because a large number of variables do not significantly improve the predictive accuracy, using only these eight essential factors can be a more effective strategy.

To enhance the interpretability of the prediction model, we applied the DeepExplainer method of SHAP to analyze the contribution of each variable. Using this approach, we confirmed that factors such as gestational age, birth weight, and total oxygen therapy had the greatest impact on predicting the occurrence of ROP. These analysis results not only demonstrate the model's high predictive accuracy but also provide a clear explanation of the importance of each variable. This plays a crucial role in helping medical professionals interpret and trust the prediction results in clinical settings, suggesting that AI-based prediction models can serve as transparent decision-making tools rather than as "black boxes."

However, this study had some limitations. First, about one-third of infants were excluded due to missing outcome data, and included cases showed slightly higher birth weight and maturity, which may limit generalizability to more vulnerable subgroups. Second, the KNN database did not include certain ROP-specific variables such as zone or plus disease, limiting characterization of disease severity. Third, treatment criteria may vary across institutions, which could introduce inter-center variability. Fourth, neonatal practice may have evolved over the 10-year study period, raising the possibility of temporal drift. Lastly, intervention variables were recorded only as cumulative values without time-resolved timestamps, leaving some uncertainty regarding their temporal relation to ROP onset.

Conclusion

In this study, we confirmed that ML-based models (MLP and NODE) show high accuracy in predicting the severity of ROP and in selecting high-risk groups requiring treatment. An AI model that can predict ROP using only clinical data, without retinal images, is a cost-effective and highly scalable solution that can be used as an effective early screening tool, particularly in medical environments where ROP experts are lacking. Further research is required to improve the integration of retinal imaging data with AI models to enable more precise predictions.

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Statement of Ethics

This study protocol was reviewed and approved by the Institutional Review Board of GangNeung Asan Hospital (IRB No. 2021-12-014). In addition, the Institutional Review Board of each participating hospital in the KNN approved the data registration and participation. Written informed consent for data collection and research use was obtained from the parents or legal guardians of all participants at the time of enrollment in the KNN registry.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

J.K.H., H.-K.P., T.H.K., and H.J. conceptualized, designed, and supervised the study. H.J.D., S.H.O. and S.H.K. were involved in the data collection and initial statistical analysis. T.H.K., D.J., and D.K. performed the computational analyses, developed the algorithms, and interpreted the data. J.K.H., H.-K.P., D.J., and H.J. wrote and edited the manuscript. All authors have reviewed and approved the revised manuscript.

Data Availability Statement

The dataset generated and analyzed in this study is not publicly available due to the research policy of the Korea Centers for Disease Control and Prevention. The datasets are available from the corresponding author upon request.

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